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ORIGINAL ARTICLE

Sticking, leaking, flowing: patient perspectives and methods for exploring data uses in the public-private datafied health infrastructure

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Abstract

As healthcare becomes increasingly datafied, public healthcare systems are becoming more deeply entangled with commercial digital infrastructures. While existing research has examined the political, ethical and infrastructural implications of these developments, less attention has been paid to how patients themselves experience and evaluate health data flows. This article investigates how Danish patients living with chronic illness understand data movement within public-private health infrastructures and explores methodological approaches for eliciting such perspectives. Drawing on five workshops with patients and informed by infrastructural analyses of mobile health apps, we combined written statements, an "attitude compass," and appscapes (visualisations of app data flows) to facilitate sharing of lived experience, evaluative reasoning and speculation. Analytically, we mobilise the concepts of sticky, leaky and flowing data to characterise patient experiences. Participants frequently described data as "sticky," becoming stranded across devices, institutions and systems, thereby preventing meaningful circulation to healthcare professionals. Concerns about "leaky" data emerged primarily in relation to speculative reflections on commercial third-party access enabled by digital infrastructures. The findings suggest that contextual legitimacy weighs more heavily than abstract concerns about privacy or public value: When data sharing is perceived as meaningful and potentially life-saving, itis understood as necessary and legitimate.

Keywords: Digital health infrastructure · Datafication · mHealth · Patient perspectives · Workshop methods · Appscapes · Public-private infrastructures · Critical data studies

Extended author information available on the last page of the article

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Introduction

Health is one of the commercial battlefields for the global tech and data industries. The commercial datafication of health (Ruckenstein and Schüll 2017) is manifest in dedicated apps for health and wellbeing that come prepackaged on most phones. With the Covid-19 pandemic, Apple and Alphabet came forward as central infrastructure providers for contact tracing (Oliver et al. 2020; Sharon 2020), and Palantir - originally a security tech company - partnered with UK and US public health agencies for data-intensive monitoring of the spread of covid (Lanzing 2023). The potentials and pitfalls of data-driven innovations in healthcare are a core concern for public healthcare. This is manifested in the European Health Data Space, envisioned to promote better care alongside innovations, policy development and research (Gross and Geiger 2023). This thirst for data-driven innovations is often seen as conditioned on public-private partnerships, with the prospect of private and commercial actors to gain access and create value out of public health data, including citizens' personal data. Analyzing and responding to such developments, critical research has pointed to the sphere transgressions (Sharon and Gellert 2023) of technology companies into the public health domain, and to the possible erosion of public values in healthcare and beyond that may follow from public-private collaborations (e.g. Bak et al. 2025; Gross and Geiger 2023; Prainsack and El-Sayed 2023; Sharon 2020; Taylor 2021).

Private actors not only play a role in public healthcare through developing and bringing new technologies for health to the world to combat diseases and empower patients. They also provide digital infrastructure and systems at the day-to-day backbone of healthcare practice. From the perspective of material infrastructure research (Helles et al. 2020; Flensburg and Lai 2024; Pybus et al. 2025), our own previous work on app data flows has demonstrated how public apps become dependent on commercial providers of software through the app infrastructure (Lomborg et al. 2024). A key question emerges from this literature about health datafication and digital infrastructure: Who gets to benefit from datafication and data sharing in digital infrastructures for health, and for whom should it create value? We suggest that addressing this question requires an understanding of how patients themselves understand health data flows in these infrastructures. In particular, it requires methodological approaches capable of rendering largely invisible data infrastructures visible, so that they can be reflected upon, questioned, and debated. We therefore examine both patients' experiences of health data flows and the methodological conditions under which patient understandings of the infrastructures for datafication of health can be articulated. Accordingly, this paper addresses two questions: How do patients with chronic illness in Denmark experience and evaluate the movement of digital health infrastructures? And how can workshop-based methods be used to cultivate patients' deliberation and patient perspectives concerning legitimate and meaningful data flows within digital infrastructures?

We approach these questions by through an empirical study with patients on the perceived obstacles and benefits of data sharing grounded in the mHealth apps they use as part of their everyday living with chronic disease in Denmark, one of the most digital and datafied welfare states in the world. In this study we bridge infrastructural and user-centric perspectives on datafication by showcasing how we can link



infrastructural analysis of digital tracking and datafication to qualitative insights on what and who patients might think infrastructures for datafication are for. By characterising patient experiences, evaluations and speculations in relation to the datafied healthcare infrastructure (Klausen 2024; Hoeyer and Langstrup 2021), we further discuss how public-private health infrastructures for datafication might be envisioned in terms of weighing concerns for the individual citizen, commercial gain, and public health. Finally, we offer methodological pointers for using infrastructural maps of app data flows, in the form of app-scapes (Lai and Flensburg 2020) in research about datafication, and for probing the commercial provision of components of digital infrastructures of the public healthcare system in Denmark.

Setting the scene: public-private entanglements in digital infrastructures

In healthcare, digital infrastructure projects are historically envisioned to provide a seamless experience for the patient when traversing various entry points of the public healthcare system. For example, electronic health records, often powered by commercial technology providers, have for decades been seen as a solution to improving care by allowing general health practitioners and public hospitals (and different units at public hospitals) to share data about the patient to ensure proper care arrangements and reduce labour (Hansen and Zethsen 2018). Moreover, as also demonstrated with the vast and varied health apps available for consumers, publicly funded as well as commercial mHealth solutions are promoted by policy makers, healthcare professionals, the tech sector and consumers alike as means to easy and on-the-fly monitoring of health indicators that are seen to empower consumer-patients to participate in their own treatment (Lupton 2013, 2019). Commercial providers thus play a significant role in building, maintaining and innovating digital and datafying infrastructures for public healthcare, and by extension for supporting and improving the welfare of citizens. At the same time, such public-private entanglements in innovation come with challenges both on the ground, in day-to-day patient monitoring and healthcare practices, and at the level of the technological backbone, where critical data studies scholars argue that commercial technology provision might be at odds with public values underpinning public institutions (Taylor 2021). Such critique is echoed in some political economic analyses of digital transformations of welfare states, arguing that the increasing reliance on private tech companies in providing basic digital infrastructure may lead to a retrenchment in states' capacities to set their own terms of operating (Collington 2022).

Previous research on the use of health apps and datafied health shows that the promises of data-driven healthcare do not always sit well with the contextual realities of healthcare practices and the capabilities of the people - professionals as well as patients - implicated (e.g. Hoeyer 2023; Klausen and Lomborg, 2025; Lomborg et al. 2020). This can create tensions regarding the perceived legitimacy of health data collection and uses, new types of data work (Bossen et al. 2019; Lomborg et al. 2024), destabilise responsibilities and expertise (e.g., Hoeyer and Langstrup 2021), and threaten public values of equality and solidarity that are core to European welfare



societies (Bak et al. 2025; Prainsack and El-Sayed 2023; Sharon and Gellert 2023). Research on health data governance further suggests we need to pay close attention to both the political and ethical implications of datafying health and to pushing data-driven innovations with citizens' health data (Gross and Geiger 2023; Reutter and Åm 2024), not least when this push involves commercial partners (Hoeyer and Langstrup 2021; Sharon 2020).

In addressing who gets to benefit from datafication in digital infrastructures for health, and for whom it should create values, we advance from Tamar Sharon's notion of sphere transgressions, which she developed over a set of publications (2020, 2021; Sharon and Gellert 2023) to conceptualise tech corporations' encroachment into multiple spheres of social life. Sphere transgressions denote those processes where "the (legitimate) advantage these actors have accrued in the sphere of the production of digital goods provides them with (illegitimate) access" (2020, p. 1) to e.g. the spheres of health and medicine and the sphere of politics. We contend that spheres are transgressed, e.g., through digital infrastructures that enable data to flow between public and private healthcare actors, and by extension between a sphere, public healthcare, governed by welfare state principles, and a private technology sector governed by commercial logics and values. In this article, we analyse how Danish patients perceive and experience data flows in digital infrastructures and between actors within the domain of health. We offer a complementary analytical perspective to that of sphere transgressions focusing not on the transgressing actors and technologies, but on what such transgressions might feel like from a patient perspective.

The project

The present study is a part of the [name] project, which combines the charting of digital infrastructures for datafication with qualitative research on how people experience and come to make sense of ordinary life as they are being tracked and track themselves across personal, work and institutional contexts. The project followed 70 Danes from all walks of life for a year, and elicited their experiences and reflections on tracking in getting things done and leading a good life through repeated interviews and photo diaries. In total, 160 interviews were carried out. Concluding the year-long fieldwork, in 2023 and 2024, participants were invited to participate in a workshop to facilitate collective speculation and deliberation on datafied futures with like-minded participants about a specific area of inquiry. This paper focuses a subset of the study participants who were originally recruited for the study because they live with a chronic disease and in that capacity track their health, either as part of a structured and coerced procedure (Lupton 2014) administered by healthcare professionals or on their own on voluntary terms (or both). This subsample is demographically diverse and includes Danes living with heart problems, inflammatory bowel disease and diabetes.

As part of the study, participants donated data on their installed apps and app uses. These donations were used by the research team to conduct infrastructural analyses of how the health apps in the sample link to third-parties in the mobile data ecosystem and make visualisations of dataflows, "appscapes" (Lai and Flensburg 2020) which



together with written statements and an "attitude compass" served as conversational prompts used in a series of workshops at the end of the year-long fieldwork.

Exploring patient perspectives on data infrastructures through workshops

Between 2022 and 2023 we held five workshops with 24 patients (16 male, eight female) 22 to 80 years of age, the majority falling between 50 and 70 years. Two workshops were arranged in collaboration with The Danish Heart Association (Hjerteforeningen). Eight men and two women, all heart patients (some of them living with multimorbidity) participated. In addition, we held one online workshop with diabetics, in which three patients, one male and two female, took part. The final two workshops had five and six patient participants, respectively, with a balanced gender distribution. The participants in these two workshops had chronic intestinal disease or one or multiple coronary diseases. Participants were recruited either through the Danish Heart Association or through Bispebjerg Hospital, where they were informed about the project by healthcare professionals and facilitated contact with the research project. The workshops lasted between two and three hours and were facilitated by the authors.

Inspired by speculative design methods the workshops aimed to engage patients in envisioning a good datafied life (Susser 2022) from the perspective of a public-private healthcare infrastructure, and to cultivate conversations about legitimate vs. illegitimate uses of data as well as realistic vs. unrealistic expectations to what patients can and cannot be asked to track in their everyday life. To ignite these conversations, we made use of three types of conversational prompts: (1) Written statements about health data and potential uses hereof, (2) An "attitude compass" capturing relations between unfair vs. fair usage of data and between what is realistic vs. unrealistic for patients to track in the everyday and (3) Appscapes, visualisations of health data flows between private and public actors.

Written statements: We presented the participants with *written statements* about health data and the potential uses hereof and asked them to elaborate on which statement(s) they felt aligned best with their feelings or points of view. Example: 1. "The healthcare professional uses my data to optimise treatment. The tracking benefits me"; 2. "I'm unsure about whether the healthcare professional uses my data or not", and 3. "I do not think the healthcare professional uses my data and have no experience with data playing a part in my treatment" The written statements anchored participants in their lived experiences of data use within healthcare systems. ***Attitude compass***: We presented the participants with what we refer to as an *attitude compass*, a tool we had designed to capture the relation between what is unfair versus fair usage of health data (the X-axis), and what is realistic versus unrealistic to ask of patients to track in the everyday (the Y-axis). An example is location tracking, which we will be elaborated below. The attitude compass encouraged evaluative reasoning about legitimacy, fairness and purpose. ***Appscapes***: We introduced the participants to appscape visualisations of data infrastructures (Lai and Flensburg 2020) based on public health sector apps apps like MinSP [MyChart] to third parties as well as from



commercial/privately owned health apps like Clue (menstrual tracking), and Sundhedshjælp [Health assistance], the latter owned by a Danish insurance company. All the apps used in the appscape visualisations were apps that had been downloaded by the project participants, and thus formed central elements in their everyday health datafication as explored in the project. As research on data infrastructure has shown (e.g. Pybus & Mir, 2025), any app is a potential tracking device that involves multiple stakeholders in datafying the user's health: the app owner may harvest data through the app, but since key elements of the app infrastructure, the so-called Software Development Kits (SDKs) are often provided by third-party actors, these might also have access to important user data. SDKs are small building blocks of code that make apps function reliably in the first place, but they are also a possible backdoor for digital tracking, data collection and profiling through ordinary citizens' app usage (Lomborg et al. 2024; Binns et al. 2018; Pybus and Coté 2024).

Figure 1 visualises the data flows enabled through the SDKs identified in our sample of mHealth apps from participants. The apps chosen here and displayed are (1) apps endorsed and used by the public healthcare sector as part of the day-to-day monitoring of patients' health, and from which healthcare professionals in some instances seek to integrate insights with other patient data to improve treatment and care, (2) apps which a patient needs in order to access public healthcare services (Fig. 2), and (3) commercial health apps that participants themselves use in their personal lives without necessarily linking them to public healthcare. These apps are

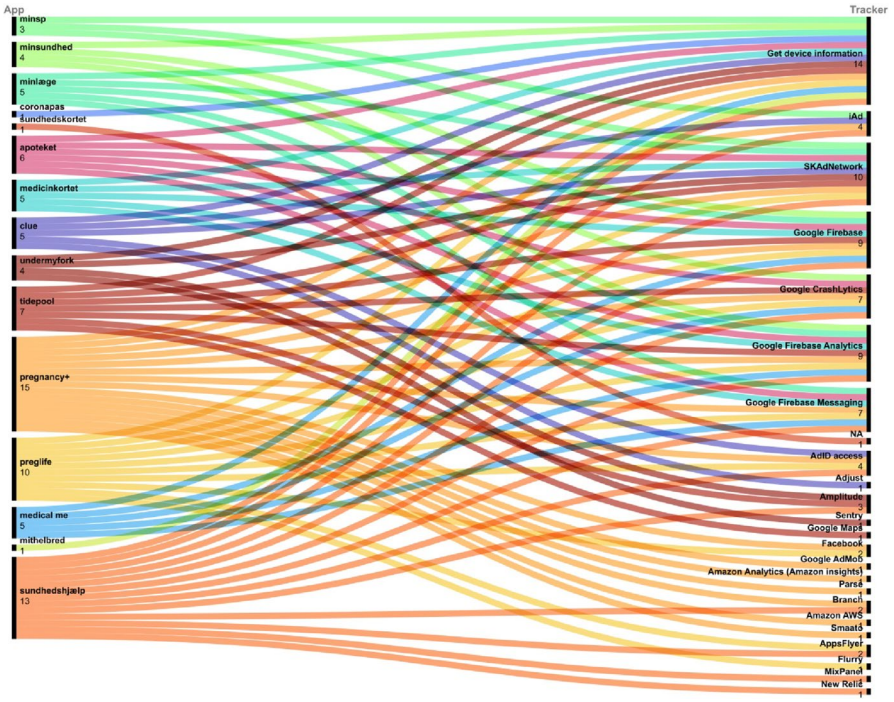


Fig. 1 Appscape of data flow from health apps

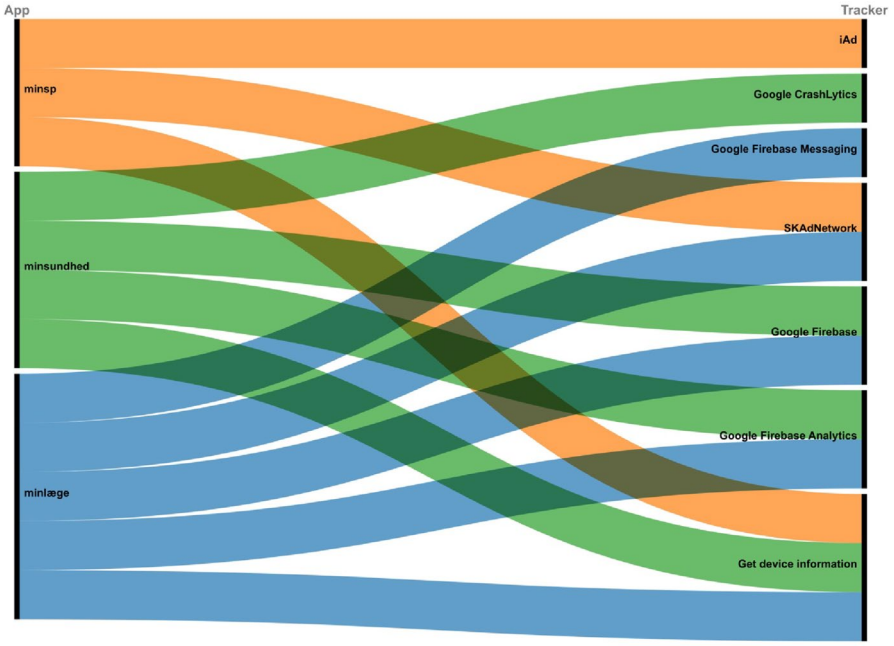


Fig. 2 Data flow from apps needed by patients to access public healthcare services

then, following Lupton's (2014) taxonomy, either pushed or coerced upon patients, or voluntarily taken up.

Figure 1 specifically displays how third parties are linked to specific apps located on a user's phone via SDKs as infrastructural building blocks in the app code. Such third-party services are embedded in apps for various purposes including analytics, functionality, interoperability and profiling. The integration of third-parties in apps displayed in Fig. 1 shows that while commercial apps may involve more, and more diverse third-parties through SDKs, private third-party actors also figure in public apps. Among the commercial third-parties are standard SDKs integrating the app with the iOS or Android platforms, owned by Apple and Alphabet, and SDKs used for tracking for example location and targeted advertising, owned by big tech companies as well as smaller businesses in the mobile data market. As will be evident from the analysis, the app infrastructures and integrated SDKs in Fig. 1 were presented as possible channels for the flow of patient data, and invitations for further inquiry and speculation in the workshops about how data might move across actors, systems and sectors via mobile apps.

Appscape analysis may itself be considered a methodological contribution to charting key infrastructural tenets of mHealth datafication (Lai and Flensburg 2020). However, in this study, we use appscapes as a data collection *prompt* for workshops. In that sense, the appscapes do not simply represent actual data flows; they actively prompt the formation of patient-driven understandings about how data moves, what it does and where it ideally should and should not flow. The appscape thus made an otherwise invisible infrastructure (partially) visible and afforded intersections between



different types of knowledge, e.g. the knowledge we have as researchers about datafication infrastructures and the knowledge the participants have about life with chronic illness as recipients of care.

Summing up, the three workshop prompts staged different registers and encounters with data and infrastructures, each contributing distinct elements to how participants experience, evaluate, imagine and speculate about health datafication and public -private entanglements in digital infrastructures.

Analysis of workshops: sticky, leaky and flowy data

The workshops were recorded and transcribed. The transcripts were then coded in NVivo using thematic coding (Braun and Clarke 2006). Both authors carried out an initial open coding followed by a discussion of the emerging patterns in participants' descriptions of data movement, breakdowns and concerns. Subsequently, the concepts of "sticky" and "leaky" data, drawing on Hoeyer (2023), were used as sensitising concepts to organise and refine these patterns. Other themes considered during our discussion of the open coding included "trust", "responsibility" and "data accuracy". While these were analytically relevant, they were ultimately integrated into the broader framework of stickiness and leakiness to maintain conceptual coherence.

The themes "sticky" and "leaky" data are inspired by Klaus Hoeyer (2023, p. 61) who mobilises stickiness and leakiness in relation to datafied healthcare. The concepts were originally used by John S. Brown and Paul Duguid in *The Social Life of Information* (2000). Here they discuss the effects of digitalization on information and how digital information tends to be "too sticky" or "too leaky". "Sticky" refers to when data is stuck to one actor or entity within the infrastructure (e.g. patients not being able to share tracking data or healthcare professionals not being able to share patient data with other public sector actors)."Leaky" on the other hand refers to uncontrollable data flows, where data is circulated to potential unknown third parties and used for unknown purposes. Between sticky and leaky, lies a desirable middle ground, where data are neither too sticky or leaky, but in a controlled manner flows in between or stays with known actors and are used for clearly stipulated purposes.

In the following we present an analysis of the empirical material from the workshops. The findings shared and discussed focus on the patient voice situated within a digital "care infrastructure" (Danholt and Langstrup 2012; Lomborg et al. 2021) which comprises both private and public actors. The analysis is organised in four sections, each of which highlights recurrent themes from across the workshops. We unpack these themes in relation to the different conversation prompts used: Attitude compass ("Legitimate and illegitimate purposes for tracking" and "Sticky and stranded data"), written statements ("Sharing with whom?") and the appscape ("Leaky data"). Taken together the four sections highlight the contrasts in the empirical material.



Legitimate and illegitimate purposes for tracking – eliciting everyday experiences through the attitude compass

As described above, the attitude compass was introduced as a conversation prompt to chart participants' sense of legitimate versus illegitimate usage of data and realistic versus unrealistic expectations to what patients can track and share with healthcare professionals in their everyday life. The participants were handed pen and sticky notes and asked to write down their ideas to what kind of patient data they could imagine tracked for healthcare purposes and then deliberate on each specific kind of data mapping the post-its onto the attitude compass. This was to bring perceptions of what patients could realistically be asked to track into dialogue with issues of legitimate vs. illegitimate tracking.

In this intersection patients negotiated weight, blood pressure, pulse and sleep. The first three mentioned were by heart patients perceived as both realistic for patients to track in their everyday as well as fair, due to the purpose this tracking data would serve in relation to their heart disease. Sleep was considered realistic to track but illegitimate: The data was imagined to be of potential relevance in relation to treatment but sleep data was considered too private and too irrelevant: "It's nobody's business how I've slept […] I'm probably a little like "Big Brother is watching you", like, it has to make sense, it has to serve a purpose. It should not just be like "come everyone and learn all kinds of stuff about me" (Patient six, male, workshop five). This attitude towards sharing sleep tracking data was followed up by another participant: "The healthcare professional does not necessarily need to know how I've slept the last fourteen days, not if it has nothing to do with anything. I simply do not think so" (Patient two, female, workshop five).

Other data types were collectively ruled out as unreasonable. For example, in one workshop heart patient participants agreed that even if possibly realistic and relevant, sexual habits and mental well-being would never be a fair thing to ask patients to track digitally. These areas of life and body were considered too personal and the participants had difficulties imagining situations where it could be vital for diagnosing or treatment that the healthcare professional had insight into these data. Location tracking was evaluated differently in different workshops. Several of the heart patients found it not only legitimate but also necessary and comforting to have one's location tracked automatically in the case of a cardiac arrest:

Patient two: "Location can be vital."

Researcher: "So sharing location is ok?"

Patient two: "Yes, if I drop dead […] Speed [of receiving medical aid] is important. When you have a critical illness, it provides a lot of safety knowing that someone knows where you are" (male, workshop three).

Patients living with intestinal disease on the other hand preferred not sharing location when using apps in relation to illness and treatment. "If I had an illness which might suddenly hit and cause me to drop dead, I might also say yes to having my location tracked - but I don't […] I prefer to not pass around information about where I am. I also don't think it's particularly relevant for my illness" (Patient four, male, workshop three). Location tracking in relation to illness was discussed as a contextually sensitive issue: It is not always needed and as such perceived of as legitimate and



meaningful, but for heart patients sharing location might mean a difference between living and dying and for this reason this patient group found it purposeful and fair to have their location automatically tracked.

Using the attitude compass in the workshops we found that patients' perceptions of (il)legitimate purposes for tracking were mobilised with reference to whether or not the activity being tracked was considered purposeful in relation to their specific case and illness(es). This is similar to what other work around data-tracking has found (Kennedy et al. 2020; see also Nissenbaum 2010). As such context sensitivity emerged as a headline in patient perspectives on fair vs. unfair tracking. This points towards a need for transparency and explicitness in the communication to patients about what they should be tracking and sharing and why it is purposeful in relation to illness to track a specific activity. Concerns about privacy or public value were thus overshadowed by the question of contextual legitimacy of tracking as location in the most literal sense can be "vital" as pointed out above.

Sticky and stranded data - attitude compass and experiences of undesirable data movements

In the discussions about (il)legitimacy and what was thought of as realistic for patients to track, the participants also described experiences relating to infrastructure and to data stranding somewhere in the infrastructure. Stories were shared about how data tracked on commercially developed technologies like an Apple watch or a Fitbit would "stick" to the patient in the digital care infrastructure, or as one patient humorously puts it "It's our Mickey Mouse watches against the big system" (Patient three, male, workshop two). In these experiences data are sticky and do not flow in a controlled and desired manner within the infrastructure and between patient and hospital. Instead, data strands within the commercial device or app, and does not become part of the public infrastructure. "When you produce so much data and note down so much stuff and you use the technology to upload the information, then why the heck are you sometimes meeting someone [a healthcare professional] who will just be a one big question mark? This frustrates me" (Patient three, female, workshop one). This points to how vast amounts of data is being generated through apps but not used by the healthcare professional i.e. the public health system. Patients also agreed that hospitals rely first and foremost on their own devices and measurements: "I never shared data from my watch with the hospital. They haven't asked and they measure themselves" (Patient one, male, workshop two).

Patients also shared frustrations about data sticking to the devices used in the public health sector: "I experienced it with an ECG measurement, where it was sent off into the darkness, not a damn thing happened. My GP said nothing, The Heart Association said nothing, the hospital said nothing. Then I had another ECK done and I demanded to speak to a heart doctor before leaving. I'm unsure about whether to trust the system or not." (Patient three, male, workshop two) Another patient had worn a blood pressure monitor for 24 h "and when I returned the device and measurements, they couldn't use it. The device had been too old or something" (Patient four, male, workshop two).



Situations where health data strands with the actor it has been shared with also emerged in the workshops. This is for instance the case when patients experience that their medical records cannot be shared between e.g. hospital or GP or even between different departments of the same hospital: Patients describe this through the metaphors of hospital departments being "silos" (Patient two, female, workshop one) and having "water-tight bulkheads" (Patient one, male, workshop one). Another case of data stranding relates to new healthcare professionals being assigned to a patient's case: "Sometimes when a doctor is new at the hospital - I think, all the data they've received and all they've written down, they don't use this, because often they'll ask you all kinds of questions and be like "Oh ok, so you've had nine cardiac arrests?" " (Patient one, male, workshop one). Data also strands in the public-private health ecology as a result of devices or apps not being interoperable or not working as they've promised to, which according to some participants is the case with *MinSp* [*My Chart*]: I was so excited when I saw in *Min SP* that there was a button you can push and you'll connect to your health data on your iPhone. It just doesn't work. It's there, but it doesn't work. And no one can explain why or give support to make it work" (Patient two, female, workshop five).

One singular reason for this prevention of data being used cannot be singled out. The participants considered reasons for this and pointed to the workload and lack of time for employees in the health sector as a potential explanation. Another aspect which in a patient perspective potentially is causing data to stick to the patient and their commercial device is healthcare professionals' lack of trust in the accuracy of the data provided by these technologies. "There are at least several medical doctors saying that, if we could trust those watches, and really develop it, and perhaps send all data to a central server, we could save incredible amounts of money and fears of dropping dead with atrial fibrillation, heart fibrillation and such things" (Patient three, male, workshop four). Commenting on this is patient two shares that he and his medical practitioner had in fact compared blood pressure data from his watch with pulse data from the equipment used at the hospital and found it to be "not 100% alike but at least close".

Through reflections like these, we see how the ideas about accuracy on the side both patients and healthcare professionals, make data stick to the commercial devices used by the patients i.e. not the devices used in the hospital or GPs clinic. We also see that patients imagine central nodes in the digital infrastructure, "a central server" (Patient three, male, workshop four), to be a solution to the problem of sticky and stranding data.

Sharing with whom? – evaluating data sharing through written statements

The analysis so far has introduced workshop findings about patient perspectives on (un)realistic vs. (il)legitimate tracking and sticky data. We now turn to the discussions the workshops brought about in relation to *who* patients preferred to share or not share data with. Here we present insights brought about when the workshop participants were asked to choose between written statements that best captured their



perceptions of the use of tracking data in healthcare. We asked participants to choose between (1) The healthcare professional uses my data to optimise treatment. (2) I'm unsure about whether the healthcare professional uses my data or not. And (3) I do not think the healthcare professional uses my data and have no experience with data playing a part in my treatment. Many patients choose the third statement: "I've placed myself on three, because I don't think the healthcare professionals can use those data yet, but I think they will be able to at some point. They don't have a system that can read this data in a fast and efficient manner. I think this is the problem […] They can't receive the data and they can't read the data. They can on a Holter-monitor, but the day they can get it [the data] from a watch, well this will be fantastic" (Patient three, male, workshop four). In the same workshop another participant went with three as well: "I have yet to experience that the healthcare professional uses my data, because I cannot send them to him" (Patient one, female, workshop four). Experiences with sticky data can also relate to the type of data being tracked. A patient shares how his healthcare professional is very keen on using the blood pressure data tracked on the patient's Apple watch: "We use the measurements I have on my iPhone. I send these to a central or to his server, so he has it when I come for a check-up. He's very happy about that and says that he would like to expand with other stuff [tracking data] too. But they [healthcare professionals] can't - they don't have the opportunity to read it." In keeping with the experiences of a health infrastructure not being up to speed yet another point is raised about the imagined complexities arising when different devices are plugged into the infrastructure "I think the technology is not fully developed for this yet. You might be able to send something from an Apple watch to your GP, but can you do that from a Garmin or something completely different as well? There are so many different… They can't receive [data] from so many different devices. I think there's a hurdle there too. I don't think it's due to lack of will" (Patient one, male, workshop four).

While this section focuses on participants' preferences regarding who should have access to their data, these discussions frequently overlapped with experiences of sticky and stranded data. Participants expressed a willingness to share data with healthcare professionals, but simultaneously described digital technology as an obstacle to such sharing. The overlap between the two themes therefore reflects how questions of data sharing are closely tied to questions of data movement: Data is experienced as problematic not only when they flow to unintended recipients, but also when they fail to reach the intended ones.

The patient perspectives shared here paint a different picture than glossy policy documents and techno-solutionist visions for a data-driven health sector. Instead, when introducing patients to the written statements, we learn from the workshops that data stick to bodies, devices and assumptions about measurement inaccuracy. We also learn that data strand between actors in the public-private infrastructure, being "sent off into the darkness" or not being able to flow the "Mickey Mouse watches to the big system". In particular, data from commercial devices (private tech) seems to be stuck or stranded, before it reaches the tech systems and care practices of the public health system Their experiences of data that fail to move prompted reflections on how the system is organised and how it ought to function. The written statements thus triggered reflections on the possibilities and limitations of the digital health infra-



structure and on the organisational and technical conditions required for meaningful data sharing.

Leaky data – appscapes and imaginings of datafication infrastructure

The workshops brought about discussions about the potential consequences of health data uncontrollably flowing to unknown actors outside the public-private health infrastructure. Unlike the previous sections, which primarily drew on participants' experiences and evaluations of tracking and sharing data, the discussions analysed here were prompted by the appscape visualisations and centred on speculative reflections about possible data flows, recipients and consequences of health data leaking. Especially the appscape visualisations pushed conversations where concerns about leaky data were shared. When presenting the appscape in the workshop we contextualised it through examples of how an app like *Min SP* [*MyChart*] when performing communicative tasks between patient and public health actor relies on bits and pieces placed within the privately owned app infrastructure and which allow third parties such as various SDKs from Google to be present in the app infrastructure, as visualised through the appscapes presented earlier in this article. It was characteristic across the workshops that the appscape visualisation the appscape shifted conversations from participant experience and evaluation towards more speculative registers about possible data flows, recipients, uses, and consequences. Importantly, the statements analysed in this section should not be understood as accounts of actual experiences with third-party data use. Rather, they provide insight into how participants made sense of possible infrastructural arrangements and consequences when presented with visualisations of public-private entanglements at the backend of their mHealth apps.

"Who's watching? I might get a bit conspiratorial, but I would be nervous about that those who produce the medicine I get injected into my body every week can see that the medicine is working, and because of this they would change it so it doesn't work anymore and I'll keep on buying it […] Like, "if everyone is cured, our business is dead" So are they [tech-pharma companies]" even interested in curing people or do they just want my data to fiddle with the medicine making sure that we'll never be completely cured? (Patient six, male, workshop five). This statement can be read as mobilising the "hostile worlds" doctrine, i.e. the idea that the market and other spheres of social life ought to remain separate (Sharon 2021). It also illustrates the unease that may arise when participants contemplate a digital health infrastructure in which commercial and public actors are perceived to be intertwined. When thinking out loud about a digital infrastructure where the market has transgressed into both the public sphere and the sphere of health, in what Sharon calls a "multi-sphere ontology" (2021), the patient questions motifs, values and consequences. These statements are best understood as speculative interpretations of digital health infrastructure and the role of commercial actors within it. The appscapes give rise to situated interpretations of infrastructure that are not based on direct knowledge but on attempts to render the system intelligible. With the appscape visualisation on the table the discussion continues, and a participant reacts to the information that *Sundhed.dk* [*Health.*



dk] might not be a 100% publicly contained app as data flows from it to commercial third-parties for operational purposes. "I don't think we should be making things like *Sundhed.dk*, where someone from the outside can enter… It should be the doctor and me only. I actually think of it as theft - just as if someone were to come and steal my chair. It's mine. Just because we need their [Google's] building blocks to make an app, it doesn't mean that they have the right to use or look at data or whatever. The public [authorities] should keep us safe, as they're asking us to use these apps" (Patient one, female, workshop five). The quote similarly mobilises a hostile worlds logic: Whereas the first participant questions whether commercial motives are compatible with the healthcare, the latter participant draws a clear boundary between public healthcare and commercial actors. The characterisation of third-party data access as "theft" signals a rejection of the idea that commercial actors have a legitimate claim to data generated in the context of care. In both cases, participants express unease with the blurring of boundaries between public healthcare and the market, reflecting concerns that values associated with commercial actors may conflict with those underpinning healthcare.

The appscape discussions also revealed how concerns about possible data leakage were often accompanied by expressions of resignation (Bagger et al. 2023). As part of their encounters with public healthcare, patients share data without necessarily feeling comfortable doing so or believing that they have a meaningful choice. As one participant remarked, "You'll just have to trust that the framework legislation will protect against data ending up in the wrong places. It's no use trying to prevent data from spreading" (Patient one, male, workshop one). Yet, the same participant described carefully reading consent forms before agreeing to share data, combining resignation with an attempt to maintain some degree of control. In this patient's account, responsibility for protecting personal data seems to ultimately rest with the patient himself.

The appscapes extend beyond experience and evaluation by prompting participants to reason about the workings of the digital healthcare infrastructure, thereby foregrounding the opacity and significance of infrastructural arrangements of health datafication, and resulting tensions around responsibilities for meaningful and legitimate data uses.

Discussion and concluding remarks: debating legitimate data uses

Our analysis has two main take-aways, one empirical and one methodological. In this section we highlight these contributions and point out new questions arising in this context. Debating infrastructures for datafication - whether with citizens or with the public at large - requires that we bring empirically grounded knowledge of who gets and uses which data with what purposes more systematically to the table.

First, from the workshops we learn that data oftentimes is experienced by the patients to stick. The transfer of data between devices and/or actors is not always seamless. We also learn that conversations around leaky data and third-party data use were trickier to engage in as patients here could not draw on lived experience from the everyday. Instead, patients here had to rely on speculative registers about



what data might be used for and by whom. Though being framed by a speculative mode, the issues of leaky data, black boxed recipients and third-party uses of data within the digital health infrastructure, created an uneasy sense amongst the patients. Some were more resigned and expressed that you cannot avoid your data flowing to unknown recipients. Others were negatively surprised by the knowledge they gained in the workshop about Danish public health sector apps being partly run by use of third-parties and perceived it as data theft. It is noteworthy that workshop conversations about infrastructural issues and third-party data use had a tendency to slide into the topics of surveillance and privacy. The participants clearly were more familiar with the privacy and surveillance vocabularies than with an infrastructure agenda. We find it both understandable and interesting that third-party data use does not figure prominently in our participants' imagination. We see an acute need for cultivating collective conversations about commercial data infrastructure both within health and medicine as well as in other spheres of social life. This is not to sidestep the importance of privacy and surveillance as important issues impacting the common good. However, privacy might blind us, as Sharon points out, to the broader societal harms (2020, p. 545) brought about by sphere transgressions and even pave the way for more transgressions, as big tech adapts to non-market values like privacy when justifying the Googlization of health (Sharon 2021). We thus locate a need for critical consciousness building about what is at stake for both patients, healthcare professionals, health sector and welfare states when infrastructural resources possibly enable data to leak and uses and recipients are unknown or poorly understood. The stakes are high as we repeatedly are being told that commercial data infrastructures and uses might sit uneasily with public values that our datafied welfare systems are expected to safeguard (Prainsack and El-Sayed 2023; Bak et al. 2025). With this study we follow the call for a shift from individualistic data rights towards a framework of data solidarity (Prainsack and El-Sayed 2023), where data governance is oriented toward collective benefit and meaningful control and use. The stickiness of data—whether stranded in commercial devices or siloed within institutional systems—undermines the promise of welfare-based integrated care and disproportionately burdens patients who for the moment are left to bridge these infrastructural gaps themselves. Meanwhile, the leakiness of data, especially to unknown third parties, introduces a sense of vulnerability and mistrust, challenging the notion of solidarity that underpins the provision of healthcare in the welfare state.

Methodologically, our study demonstrates the value of combining different registers when scaffolding conversations about health infrastructures: The written statements elicited experience-based accounts of data sharing, the attitude compass facilitated evaluations of legitimacy and purpose, and the appscapes encouraged speculation about data flows and infrastructural arrangements beyond participants' direct experience. Taken together, these methods enabled participants to articulate concerns, expectations and interpretations of the public-private health infrastructure from multiple perspectives. By making otherwise invisible data infrastructures partially visible and open for collective discussion, the workshops created opportunities for reflection on questions of power, responsibility, value and trust in datafied healthcare.



In this paper we have shown how infrastructural analysis can be bridged with user-centric perspectives on datafication. More empirical work is needed to further qualify the conversation about datafication, and, we will add, more attention needs to be directed towards developing methodological approaches to bringing infrastructural analysis in conversation with lived experience perspectives and debates. In this article we have amplified the patient voice and perspective as an empirically grounded contribution to the conversation about the good life in a datafied welfare state. Our analysis suggests that the question of the contextual legitimacy of tracking weighs more heavily than questions of public value or privacy: If lives are at stake, which sometimes is the case for chronically ill patients, sharing data is considered necessary, legitimate and provides a feeling of safety.

If sphere transgressions are becoming the standard solution to efficient digital infrastructure, and public values to a growing extend are adapted into big tech's justifications for encroaching into the social spheres of our lives, there is a need for carving out collective spaces and conversations about infrastructure and *how* and *why* infrastructure and ownership are affecting all of us and our opportunities to live good lives with datafication. These conversations should not replace but be held in tandem with the ones we are already having and should continue to have concerning privacy and surveillance.

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Declarations

Competing interests The authors declare no competing interests.

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